BACHELOR OF TECHNOLOGY

PROJECT REPORT

CUSTOMER CHURN ANALYSIS AND DASHBOARD

An interactive business analytics study using Excel and Python

Submitted in partial fulfilment of the requirements for the award of the degree of

Bachelor of Technology in Data Analytics

Submitted By – Sadiya

Certificate

This is to certify that the project report entitled “Customer Churn Analysis and Dashboard” is a bona fide record of work carried out by the student named on the cover page under the supervision of the project guide named there, in partial fulfilment of the requirements for the degree of Bachelor of Technology in Data Analytics at the institute named on the cover page.

The work has been reviewed to the extent required by the department. The student has demonstrated an understanding of the project objectives, data preparation, analytical methods, implementation choices, and limitations.

Declaration

I declare that the project titled “Customer Churn Analysis and Dashboard” is my academic work completed as part of the Bachelor of Technology programme at the institute named on the cover page, under the guidance of the project guide named there. This report has not been submitted, in whole or in part, for another degree or qualification.

I have acknowledged the dataset, software, documentation, and other sources used. The analysis is limited to the public IBM sample data and methods described here. Results in this report should be reconciled with the final project workbook before submission.

Acknowledgement

I would like to express my sincere gratitude to my project guide and the faculty members of my institute for their guidance and academic support. Their feedback helped shape the problem, organise the analysis, and present the work clearly.

I acknowledge IBM for making the Telco Customer Churn sample dataset available and the developers of the open-source tools used in this project. I also thank my family and classmates for their support. Any remaining errors are my responsibility.

Abstract

Customer churn is the loss of customers over a defined period. For a subscription-based telecommunications business, understanding which customer groups show higher churn can help teams investigate service experience, pricing, contract design, and retention opportunities. This project develops descriptive customer churn analysis using the Telco Customer Churn dataset and presents it through an Excel dashboard and a Python-based Streamlit application.

The work covers data inspection and cleaning, calculation of customer-level indicators, and comparisons of churn across contract type, tenure, internet service, payment method, monthly charges, senior-citizen status, partner status, and dependents status. The Excel workbook uses summary tables, PivotTables, charts, KPI cards, and slicers. The Python application uses Pandas for data handling and Matplotlib and Seaborn for visual analysis, with Streamlit as the user interface.

This project supports exploration rather than causal explanation or individual prediction. The report gives descriptive values calculated from the public IBM CSV. The delivered project workbook and application were not available for inspection, so their specific slicer and chart behaviour cannot be asserted. Ordinary charts based on static ranges may not respond to PivotTable slicers.

Keywords: customer churn, telecommunications, Excel dashboard, Streamlit, data analysis, retention.

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Abbreviations

Term	Meaning
API	Application Programming Interface
CSV	Comma-Separated Values
EDA	Exploratory Data Analysis
KPI	Key Performance Indicator
ML	Machine Learning
UI	User Interface

1 Introduction

This chapter introduces the business problem, project context, and scope.

1.1 Background

Subscription businesses depend on continuing customer relationships. A customer who ends a service relationship represents churn. Churn can affect recurring revenue, acquisition efficiency, and planning. Historical customer records can show patterns associated with churn and help teams decide where further investigation may be useful.

A dashboard makes measures easier to review than raw records. This project combines a spreadsheet dashboard with a small Python web application so that the analysis can be explored with familiar office tools and an interactive interface.

1.2 Need and project overview

An overall churn figure may hide differences between customer groups. Contract commitment, tenure, service mix, payment method, charges, and customer profile provide context for retention planning. The project prepares and explores a public telecommunications dataset, calculates descriptive summaries, and presents them visually. The observed relationships do not explain why customers leave.

1.3 Scope

- Use the Telco Customer Churn dataset as the analytical input.
- Prepare data and document quality checks and cleaning choices.
- Compare churn across selected service and customer dimensions.
- Present KPI cards and charts in Excel and a Streamlit application.
- Document implementation, results, validation, recommendations, limitations, and extensions.

1.4 Report organisation

Chapters 2–5 define the problem, data, preparation, and methods. Chapters 6–7 explain solution design and implementation. Chapters 8–10 present the result framework, validation, and recommendations. Chapters 11–12 discuss limitations and conclusion, followed by references and appendices.

2 Problem Definition and Objectives

The project converts a broad business concern into a descriptive analytics task.

2.1 Problem statement

How can customer and service records be analysed and presented so that differences in observed churn across relevant customer segments can be explored clearly? The task is to organise historical data and communicate patterns, not to determine causes or predict an individual customer's future behaviour.

2.2 Objectives

No.	Objective	Evidence
1	Inspect and prepare the dataset.	Documented checks and cleaning.
2	Calculate headline customer and churn measures.	KPI cards and metric definitions.
3	Compare churn across selected dimensions.	Charts and segment summaries.
4	Build an Excel dashboard with filters.	Workbook with charts, slicers, and KPIs.
5	Build a Python/Streamlit application.	app.py using Pandas, Matplotlib, and Seaborn.
6	Review correctness and usability.	Recorded test outcomes and limitations.

2.3 Analysis questions

- What share of records are labelled churned?
- How does observed churn vary by contract and tenure?
- How do internet service and payment method groups compare?
- How do charges and profile categories relate to the churn label?
- What business questions should be investigated next?

2.4 Stakeholders and deliverables

Intended readers are academic evaluators, the project guide, analysts, and business users. Deliverables are an Excel workbook containing cleaned data, summaries,

charts, KPI cards, and slicers; a Python Streamlit application in app.py; and this report. Business users should treat findings as exploratory and verify them against operational context.

3 Dataset and Business Context

Interpretation depends on the dataset definition, observation period, and quality checks.

3.1 Dataset description

The Telco Customer Churn dataset contains customer-level records describing telecommunications services, contract and payment arrangements, tenure, charges, and churn status. Dataset editions can differ. Record the actual file source, retrieval date, row count, and exact fields used.

Dataset detail	Project value
Dataset title	IBM Telco Customer Churn
Source / URL	IBM sample repository; see References
File name and format	Telco-Customer-Churn.csv
Date accessed	29 September 2026
Rows before cleaning	7,043
Rows after cleaning	7,043 customer records; blank TotalCharges values retained as missing for that field only
Target field	Churn (Yes/No)

3.2 Business meaning

In this report, churn means a record is labelled as a customer who left or discontinued service. Confirm the target definition and observation period from source documentation. If no precise time window is available, describe the result as the share of records labelled churned, not as a monthly or annual rate.

3.3 Analysis dimensions

Dimension	Treatment	Question
Contract	Source categories	Does churn vary by contract type?
Tenure	Months; optional bands	How does churn compare by service length?
Internet Service	Source categories	Do service groups differ?
Payment Method	Source categories	How does churn compare by

		payment?
Monthly Charges	Numeric or bands	How does churn compare across charges?
Senior Citizen	Recorded category	How does churn differ by status?
Partner	Yes/No or source categories	How does churn compare by partner status?
Dependents	Yes/No or source categories	How does churn compare by dependents?

3.4 Data dictionary guide

Field	Type	Meaning / check
customerID	Identifier	Unique customer key; exclude from averages.
Churn	Categorical target	Standardise values and verify allowed labels.
tenure	Numeric	Months; check for negative values.
Contract	Categorical	Contract type.
InternetService	Categorical	Service category and missing/none meaning.
PaymentMethod	Categorical	Recorded payment category.
MonthlyCharges	Numeric	Verify conversion and units.
SeniorCitizen	Binary/category	Check whether 0/1 encodes status.
Partner	Categorical	Partner status.
Dependents	Categorical	Dependents status.
TotalCharges	Numeric/text in some editions	Convert carefully; blanks may mean missing.

4 Data Preparation

Preparation aims to make records consistent while preserving source meaning.

4.1 Loading and inspection

Load the source file and confirm that it opens correctly. Inspect column names and types, row and column counts, sample records, missing values, and duplicate identifiers. Record these checks before editing the data so changes can be explained and reproduced.

4.2 Cleaning procedure

Check	Action	Evidence
Column names	Trim spaces; standardise only as needed.	Field list
Data types	Convert tenure and charge fields.	Types and exceptions
Missing values	Count and document treatment.	Missing-value summary
Duplicates	Check duplicate rows and IDs.	Counts and resolution
Categories	Inspect whitespace and unexpected labels.	Distinct values
Target labels	Confirm labels and denominator.	Counts and proportions
Numeric ranges	Check plausible minimum and maximum.	Range review

4.3 Invalid values and transformations

Some dataset editions store TotalCharges as text and may contain blanks. Trim whitespace, convert valid numeric strings, identify failed conversions, and decide how to treat them. Do not silently replace invalid values with zero. Record affected rows and rationale. If tenure or charges are grouped, document mutually exclusive boundaries and retain original numeric fields for calculations.

4.4 Reproducibility and privacy

Keep an untouched original and save prepared data separately. Record filenames, cleaning steps, and preparation date. Avoid including customer identifiers in

screenshots or shared reports. Use identifiers for record-level checks only when appropriate and handle customer data according to institutional and organisational privacy requirements.

5 Methodology and Metrics

The analysis uses descriptive statistics and visual comparisons across customer groups.

5.1 Workflow

Define questions; load and inspect data; clean and document fields; calculate overall and segment measures; build charts; compare Excel and Python outputs; interpret patterns cautiously; and propose follow-up actions. Figure 1.1 may show: Source data → Quality checks → Cleaning → Measures → Excel and Streamlit views → Validation → Recommendations.

5.2 Metric definitions

Measure	Definition	Interpretation
Total customers	Count of distinct customer IDs where available.	Population after documented exclusions.
Churned customers	Count labelled Yes.	Number of labelled churners.
Churn rate	Churned ÷ eligible customers × 100.	Share labelled churned; state denominator.
Average tenure	Mean tenure in months.	Average recorded service length.
Average monthly charges	Mean MonthlyCharges.	Average recurring charge; state currency.
Segment churn rate	Churned in segment ÷ all eligible in segment × 100.	Within-category rate.

5.3 Denominators and comparisons

A segment churn rate divides churned records by all eligible records in that segment. It differs from the share of all churners belonging to the segment. Titles and axes must say which measure is shown. Display customer counts with rates because small groups can produce unstable percentages. These are descriptive associations and do not establish causation.

5.4 Visual choices

- Use titles that name the dimension and metric.
- Keep churn/retained colours consistent.
- Label percentages, months, and charges with units.
- Avoid unnecessary decoration and three-dimensional charts.
- Include the dataset date or version in the final application.

6 System Design

The solution provides a spreadsheet interface and a Python web interface for the analysis.

6.1 Architecture

The logical architecture has input data, preparation, analytical summaries, and presentation. The Excel workbook stores data and summary structures. The Streamlit app loads a data file, prepares fields through Python, and renders KPIs and charts. To agree, both outputs must use the same source and equivalent cleaning and metric definitions.

Layer	Excel	Python
Input	Clean_data worksheet or imported table.	Dataset file loaded by app.py.
Preparation	Documented cleaning and PivotTable source.	Pandas conversion and cleaning.
Analysis	PivotTables, formulas, summary ranges.	Pandas grouping and aggregation.
Visuals	Charts, KPI cards, slicers.	Streamlit with Matplotlib / Seaborn.
Interaction	Slicers and worksheet navigation.	Streamlit controls, if implemented.

6.2 Dashboard design

Place a project title and refresh context at the top, KPI cards below, charts in a consistent grid, and slicers in a labelled area. Raw/clean data, PivotTables, and Dashboard sheets may be separated. State what each slicer filters and provide a clear reset or Select All path.

6.3 Slicer and chart relationship

Excel slicers filter connected PivotTables and PivotCharts. A standard chart built from a copied static range may not respond to slicer changes. Check Report Connections for every slicer and confirm that each intended chart is a PivotChart

or uses a dynamic filtered range. KPI cards based on fixed formulas may also stay unchanged; test them independently.

6.4 Streamlit layout

The application should guide users from a title and short description to summary indicators and analytical views. Keep chart labels readable and make selected populations visible. Handle missing files, required-column errors, and invalid numeric values with clear messages.

7 Implementation

This chapter describes the technologies and implementation responsibilities.

7.1 Technology stack

Technology	Role
Microsoft Excel	Data review, PivotTables / summaries, KPI cards, charts, slicers.
Python	Analysis and application logic.
Pandas	Loading, cleaning, grouping, aggregation.
Matplotlib	Chart rendering.
Seaborn	Statistical plotting and styling where used.
Streamlit	Web application interface.
app.py	Main Python application source.

7.2 Excel implementation

Organise the workbook around imported/raw data, cleaned data, a Pivot Table worksheet, and Dashboard. Create PivotTables from a consistent Excel table or source. Document summary definitions and format percentages explicitly. KPI cards may include total customers, churned customers, churn rate, average tenure, and average monthly charges. Record formulas or PivotTable links and test whether slicers update each card.

7.3 Python implementation

The app.py entry point typically imports required libraries, loads the dataset, validates required columns, prepares data types, computes overall and grouped summaries, displays KPI values, and renders charts. Keep data preparation separate from rendering where practical. Describe exact controls only after inspecting the final app.py. Insert a screenshot of the running application and record the tested launch instructions in Appendix B.

7.4 Usability and evidence

- Check expected input file and required columns.

- Show readable messages for invalid inputs.
- Match labels to dataset and report definitions.
- Keep charts legible at browser size.
- Prefer portable relative paths where practical.
- Use final screenshots that match tested results.

8 Results and Discussion

Enter verified values from the final workbook and application. No unverified numerical findings are assumed here.

8.1 Overall KPI values

KPI	Value	Definition / notes
Total customers	7,043	Distinct customer IDs in the IBM sample.
Churned customers	1,869	Count labelled Yes.
Overall churn rate	26.54%	1,869 ÷ 7,043 × 100.
Average tenure	32.37 months	Mean tenure in the sample.
Average monthly charges	$64.76	Mean; dataset charges are in US dollars.

8.2 Churn by contract

Table 8.1 gives within-category rates calculated from the IBM sample. Month-to-month customers have the highest observed rate in this descriptive comparison. Contract type is associated with churn in this dataset; this comparison alone does not show that contract type caused the outcome.

Contract category	Customer count	Churned count	Within-category rate
Month-to-month	3,875	1,655	42.71%
One year	1,473	166	11.27%
Two year	1,695	48	2.83%

8.3 Churn by tenure and service

Tenure is strongly descriptive of customer stage: the dataset mean is 32.37 months, while the segment comparisons should use the exact bands configured in the workbook. Insert or retain the workbook's tenure chart and label its categories. For Internet Service, the within-category rates are 18.96% for DSL (459 of 2,421), 41.89% for Fiber optic (1,297 of 3,096), and 7.41% for customers with no internet service (113 of 1,526). These are dataset-level comparisons, not causal effects.

8.4 Payment method and monthly charges

Within-category churn rates are 45.29% for Electronic check (1,071 of 2,365), 19.11% for Mailed check (308 of 1,612), 16.71% for Bank transfer (automatic; 258 of 1,544), and 15.24% for Credit card (automatic; 232 of 1,522). The average MonthlyCharges is $74.44 for churned records and $61.27 for retained records. Payment method and price may be associated with other factors; these summaries do not establish a cause. If the workbook uses charge bands, retain its documented cut points and show category counts.

8.5 Customer profile categories

The observed rate is 41.86% for customers coded SeniorCitizen=1 (476 of 1,142) and 23.61% for SeniorCitizen=0 (1,393 of 5,901). Customers without a recorded partner show 32.96% churn (1,200 of 3,641), compared with 19.66% for those with a partner (669 of 3,402). Customers without dependents show 31.27% churn (1,543 of 4,933), compared with 15.45% for those with dependents (326 of 2,110). These are descriptive comparisons; avoid stereotyping or using profile categories as the sole basis for decisions.

8.6 Findings summary

Area	Verified observation	Count / context	Caution
Contract	Month-to-month highest at 42.71%	3,875 customers	Association only.
Tenure	Dataset average is 32.37 months	7,043 records	Use workbook bands for graph.
Internet service	Fiber optic 41.89%; DSL 18.96%; none 7.41%	Counts shown in text	Not causal.
Payment method	Electronic check 45.29%; other methods 15.24–19.11%	Counts shown in text	May reflect other factors.
Monthly charges	Churned mean $74.44; retained $61.27	Grouped by churn label	Descriptive difference.
Customer profile	Rates vary by recorded categories	Counts shown in text	Use responsibly.

8.7 Interpretation

A dashboard can show where churn labels are more common and guide the next question. A segment with a higher observed rate may merit review of customer journey, service quality, pricing, or retention offers. Before a decision, validate the pattern over an appropriate time period, compare counts, consider other variables, and consult operational context.

9 Testing and Validation

Validation checks data, calculations, interactions, and presentation.

9.1 Test strategy

Test data loading, field validation, metric calculations, chart values, slicer behaviour, and usability. Confirm formulas with an independent calculation or manually checked sample. Record actual outcome, date, and workbook/app version. Do not mark an unperformed test as passed.

9.2 Test cases

ID	Test	Expected result	Actual status
T01	Load canonical source dataset	File and expected 21 fields are present.	Verified from IBM CSV listing.
T02	Check row/customer counts	7,043 rows and unique customer IDs.	Dataset profile: 7,043 records.
T03	Recalculate churn rate	1,869 / 7,043 reproduces 26.54%.	Calculated from published dataset totals.
T04	Compare Excel and Python	Metrics agree within rounding.	Not verifiable; project files unavailable.
T05	Change Contract slicer	Intended tables, charts, KPIs respond.	Not verifiable; workbook unavailable.
T06	Change Partner/Dependents	Connected views update.	Not verifiable; workbook unavailable.
T07	Change service/payment slicer	Connected views update.	Not verifiable; workbook unavailable.
T08	Clear filter	Returns to documented full view.	Not verifiable; workbook unavailable.
T09	Malformed app input	Clear actionable error is shown.	Not verifiable; app.py unavailable.
T10	Review final charts	Labels, units, categories readable.	Not verifiable; app/workbook screenshots unavailable.

9.3 Excel interaction validation

For every slicer, select one value and observe which PivotTables change. Confirm each associated chart changes and inspect Report Connections. If a chart remains

static, check whether it uses a static range, connects to the intended PivotTable, and shares a compatible source/cache. Test KPI cards separately because standard worksheet formulas may not respond to slicers.

9.4 Cross-platform consistency

Compare overall and segment measures between Excel and Python. Reconcile differences in excluded rows, missing-value treatment, category spelling, rounding, and denominator definitions. Document intentional differences and align outputs when claiming a shared definition.

9.5 Acceptance criteria

- All KPI values can be reproduced from documented formulas.
- Charts use the stated measure and denominator.
- Slicers filter every view they are claimed to filter.
- The app handles expected input errors clearly.
- Screenshots and reported findings match final tested versions.

10 Business Recommendations

Recommendations should match descriptive evidence and lead to validation or further investigation.

10.1 Prioritise investigation

After verifying results, identify comparatively high-rate segments with adequate customer counts. Ask relevant teams to review contact history, service incidents, pricing changes, renewal experience, and customer feedback. Treat the dashboard as a screening tool for questions, not as a complete decision rule.

10.2 Review contract and tenure journeys

If verified results differ by contract or tenure, examine onboarding, renewal, and plan-change experiences. Review whether terms are clear and support is available at key points. Evaluate any retention intervention for customer benefit and business impact.

10.3 Investigate service and payment experience

If service type or payment method is associated with churn, assess reliability, billing clarity, payment friction, and support cases. Do not assume the category itself causes churn. Test proposed changes with customer feedback and a measured pilot.

10.4 Monitor responsibly

Use consistent definitions and refresh schedules. Pair rates with counts and time windows. Protect customer data and review whether profile fields are appropriate. Avoid unfair treatment based on demographic proxies or small, unstable groups.

Action	Suggested owner	Measure of progress	Review point
Validate high-rate segments	Customer analytics / operations	Documented root-cause review	After verified segment analysis
Review onboarding	Customer experience	Feedback and process	After journey review

and renewal	team	findings	
Audit definitions and refresh	Data analyst / dashboard owner	Reconciled workbook/app values	At each data refresh
Evaluate pilot if justified	Retention / service team	Predefined outcome and comparison	After pilot evaluation

11 Limitations and Future Scope

These limits affect how confidently the findings can be applied.

11.1 Limitations

- The supplied dataset may not represent current customers or an organisation's operating context.
- Observation period and churn definition must be confirmed.
- Descriptive associations do not prove cause and effect.
- Missing values and cleaning decisions can change results.
- Small segment counts can produce unstable rates.
- Slicer behaviour depends on PivotTable, PivotChart, and cache connections.
- The local app may depend on a particular file path or library version.
- The project does not demonstrate predictive performance or intervention effectiveness.

11.2 Future scope

- Add a documented refresh process and automated quality checks.
- Create a unified model so all Excel charts and KPIs respond to filters.
- Add trends when reliable time-stamped records are available.
- Evaluate predictive models only with leakage checks, suitable validation, and fairness review.
- Use cohort or survival analysis where data supports it.
- Collect service-event and feedback data to investigate explanations.
- Deploy Streamlit with access controls and data governance.
- Test usability and accessibility with intended users.

12 Conclusion

This project demonstrates a structured way to explore customer churn records and communicate patterns through Excel and Streamlit.

The work combines data quality review, metric definition, segment comparisons, and visual presentation. Its practical value depends on consistent data preparation and denominators, reconciliation between workbook and app, and verification that slicers control their intended views. Final numerical findings are left for entry from the verified dashboard.

The analysis is a starting point for business investigation. It can help teams ask where churn is more common in the observed records, but additional evidence is needed to explain why and whether a proposed intervention improves customer outcomes. With verified findings, clear documentation, and reliable filter behaviour, the project provides a useful academic demonstration of applied data analytics.

References

Replace or supplement these entries with the exact pages and software versions used. Include access dates and follow the department's citation style.

1. IBM. Telco Customer Churn sample dataset, CSV. https://raw.githubusercontent.com/IBM/telco-customer-churn-on-icp4d/master/data/Telco-Customer-Churn.csv (accessed 29 September 2026). Counts and descriptive rates in Chapter 8 are calculated from this file; results may differ if the project workbook uses a different edition or cleaning rule.
2. The pandas development team. pandas documentation. https://pandas.pydata.org/docs/
3. Streamlit, Inc. Streamlit documentation. https://docs.streamlit.io/
4. Matplotlib Development Team. Matplotlib documentation. https://matplotlib.org/stable/
5. Waskom, M. L. Seaborn: statistical data visualization. Journal of Open Source Software, 6(60), 3021, 2021. https://doi.org/10.21105/joss.03021
6. Microsoft Support. PivotTables, PivotCharts, and slicers in Excel. Consult the Microsoft Support documentation applicable to the installed Excel version.

Appendix A Data Dictionary and Calculation Notes

Use this appendix to make the analysis reproducible.

A.1 Field verification

- Copy exact field names and types from source.
- Record units, categories, missing counts, and cleaning actions.
- Confirm churn definition and time period.
- Record customer ID uniqueness.
- Document treatment of invalid numeric fields.

A.2 Calculation notes

Calculation	Documentation
Overall churn rate	Churn Yes / eligible population × 100; state exclusions.
Segment churn rate	Churn Yes in segment / all eligible in segment × 100.
Average tenure	Mean; document missing-value handling.
Average monthly charges	Mean; state currency/unit and missing handling.
Rounding	Display percentages to one decimal place; use unrounded values in calculations.

A.3 Figure notes

The report describes the project workflow, dashboard architecture, and chart views in text. Dashboard and application screenshots were not included because the source workbook and app were not available in the workspace. The numerical results in Chapter 8 are sourced to the public IBM CSV and should be reconciled with the student project files if those files use a different data version or transformation.

Appendix B User Guide

This guide describes the intended workbook and application workflow. The project workbook and source code were not available for direct inspection.

B.1 Excel dashboard

- Open the final workbook and review KPI definitions.
- Use slicers to choose categories; use clear-filter controls to restore the full population.
- Observe which charts and KPIs update and report any unexpected static view.
- Use the Pivot Table worksheet to review underlying summaries.

B.2 Streamlit application

The project description identifies a Streamlit application in app.py, using Pandas, Matplotlib, and Seaborn. Exact installation and launch steps depend on the project configuration and could not be verified because app.py and its dependency file were not present.

Setup detail	Project record
Python version	Not recorded in available project materials.
Dependency file	Not available for inspection.
Dataset location	Defined by app.py configuration; source not available.
Launch command tested	Not available for verification.
Known setup issue	Not assessed because app.py was unavailable.

B.3 Refresh checklist

- Obtain an authorised current dataset and record its date.
- Confirm schema and category values.
- Refresh cleaning and summaries.
- Reconcile Excel and Python measures.
- Test every slicer and application control.

- Update screenshots, findings, and report values.

Appendix C Project Viva Explanation

Prepare an accurate explanation of your own contribution and the evidence behind the results.

C.1 Short explanation

"My project analyses customer churn using the Telco Customer Churn dataset. I prepared and checked the customer records, calculated overall and segment-level measures, and presented the analysis in an Excel dashboard and a Streamlit application. The dashboard compares churn across contract, tenure, internet service, payment method, monthly charges, senior-citizen status, partner status, and dependents. The analysis is descriptive: it identifies patterns but does not prove causes or predict future churn. I validated the calculations and filter behaviour and documented limitations and possible improvements."

C.2 Viva questions

Question	Points to cover
What is churn?	Use dataset target definition and state its limits.
How is churn rate calculated?	Churned eligible customers ÷ all eligible × 100.
Why Excel and Python?	Excel for familiar filtering; Python for repeatable processing and app presentation.
What does Pandas do?	Loads, cleans, groups, and summarizes tabular data.
Why may a chart ignore a slicer?	Static range or missing PivotTable connection.
Do results prove why customers leave?	No; comparisons are associations.
What would you improve?	Refresh, filter-linked KPIs, trends, user testing, careful predictive work.

C.3 Project contribution summary

The documented project scope includes preparation of customer records, descriptive analysis, an Excel dashboard, and a Python application. A student submitting this work should describe their own contribution accurately during

assessment and distinguish their work from any supplied examples or shared materials.